

Global CO₂ Emissions Tracker by Sector

Introduction

Climate change is a pressing global issue, and tracking carbon dioxide (CO₂) emissions is essential for informed policy-making and sustainable development. This project analyses multi-year CO₂ emissions across countries and sectors, providing insights into total emissions, per-capita trends, and top polluters.

Abstract

This project presents an interactive dashboard for tracking global CO₂ emissions by sector, including cement, coal, oil, gas, and other sources. The dashboard allows visualization of emissions per country, per sector, and per capita, enabling identification of top contributors and global trends. Key performance indicators (KPIs) include total CO₂ emissions, coal CO₂ emissions, and country-wise comparisons.

Tools Used

- **Data Preparation & Cleaning:** Python (Pandas, NumPy)
 - **Visualization & Dashboard:** Power BI
 - **Dataset:** Global CO₂ emissions data from Our World in Data (OWID)
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Steps Involved in Building the Project

1. Data Collection

- Downloaded multi-year CO₂ emissions dataset from OWID.

2. Data Cleaning & Handling Missing Values

- Filled numeric columns with median values.
- Filled categorical columns (e.g., country names) with mode values.
- Dropped unnecessary columns with excessive missing data.

3. Derived Metrics Calculation

- Calculated **per-capita CO₂** and **per-GDP CO₂** metrics.
- Converted wide-format sector columns (`cement_co2`, `coal_co2`, `oil_co2`, `gas_co2`) into long format with `sector` and `co2_emissions` columns.

4. Dashboard Creation in Power BI

- **Slicers:** Country & Year for interactive filtering.
- **World Map (Choropleth):** CO₂ emissions per capita by country.
- **Treemap:** Share of global CO₂ emission by country.
- **Pie Chart:** CO₂ by country for quick comparison.
- **KPI Cards:** Total CO₂ emissions, Coal CO₂ emissions, Total emissions.

5. Testing & Validation

- Verified calculations and interactions between slicers and visuals.

Conclusion

The Global CO₂ Emissions Tracker provides a clear and interactive way to monitor and analyze CO₂ emissions by country and sector. It allows policymakers, researchers, and the public to identify top polluters, observe emission trends over time, and evaluate the impact of different sectors on global carbon output. This dashboard supports data-driven decision-making for climate action initiatives.